KAREN NGUGI

AI/ML Engineer

Personal Statement

From the invention of the camera to the rise of virtual reality, history shows how technology keeps reshaping the world of art. Karen views tech not as a passing trend, but as a force that unlocks entirely new ways to imagine, express, and connect. How exciting it is to be a software engineer because creativity is not limited by medium, but expanded by the possibilities technology brings.

Links/Websites

- LinkedIn
- GitHub

Education

AkiraChix
codeHive - Diploma in Information Technology
February 2025 – present

Skills

- **Libraries & Frameworks:** Pandas, NumPy, Scikit-learn, Model evaluation, SpaCy,DuckDuckGo_
- Search, OpenCV, Matplotlib, Seaborn, Plotly, TensorFlow, HuggingFace Transformers, Django REST Framework.
- **Databases & Vector Search:** Pgvector, Supabase, ChromaDB
- **Data Engineering:** Feature Engineering, Text Preprocessing, embeddings, pipelines
- **Cloud & Deployment:** Google Cloud Run, API hosting

References

Linda Kamau
Founder and Executive Director
AkiraChix
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Synthia Hunter Achieng
Software Engineer
OpenFunction Group
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Sharon Obanda
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Projects

1. **Tesfa: Health Risk Predicting Agent**

Developed Tesfa, an agentic AI-based tool, to address the critical gap where long-term health effects in post-war regions are often overlooked due to the urgent focus on immediate crises. Tesfa helps NGOs and health agencies shift beyond short-term relief by guiding strategic planning through tasks for sustained health interventions.

Approach and Key Learnings:

- Crawled and transformed 700+ unstructured war and health reports into high-quality data, boosting AI agent analysis of post-conflict health impacts by switching from sentence-based to semantic chunking for better retrieval accuracy.
- Embedded war and health text using SentenceTransformer models, storing embeddings initially in ChromaDB and later migrating to pgvector enabling faster context retrieval for scalable health risk assessments.
- Used Gemini-2.0-flash to structure BioGPT outputs into standardized JSON formats for precise health risk assessments in post-conflict regions, combining vector search with instruction-based and few-shot prompting to reduce model hallucinations and improve prediction reliability.
- Designed a hybrid retrieval system integrating Supabase pgvector and DuckDuckGo library, enabling robust context gathering for the Retrieval-Augmented Generation agent to deliver accurate health risk predictions in conflict zones.

2. **Breast Tumor Detection**

Developed a deep learning model to accurately classify breast tumor types from ultrasound images, employing advanced preprocessing and optimization techniques.

Approach and Key Learnings:

- Built and trained a CNN in PyTorch to classify breast ultrasound images (normal, benign, malignant) with 75% accuracy, improving data quality by removing corrupted images using PIL.
- Applied image augmentation techniques, including random horizontal flips and rotations, to enhance the diversity of breast ultrasound datasets, improving CNN model generalization.
- Optimized training with the Adam algorithm for 10 epochs with a batch size of 16, reducing training time significantly via GPU acceleration.
- Evaluated model performance with confusion matrices and classification reports using scikit-learn and seaborn, enhancing diagnostic accuracy for the ultrasound images.

3. **Housing Prices Prediction**

Created a comprehensive machine learning pipeline to forecast housing values by preparing and analyzing data from diverse sources, uncovering meaningful patterns.

Approach and Key Learnings:

- Processed and cleaned Ames housing dataset with 1,460 records using Pandas, imputing missing data with median/mode, ensuring a robust dataset for accurate price prediction.
- Conducted correlation analysis with Seaborn heatmaps to identify and drop highly correlated features (>0.8), enhancing model stability and reducing overfitting by 10%.
- Trained Linear Regression and KNN Regression models using scikit-learn, achieving an average R^2 of 80% in cross-validation for precise house price predictions, with KNN Regression consistently outperforming Linear Regression in both R^2 and RMSE metrics.